EPIK8s at BTF

From Process Variables to Operational Interfaces

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Doctoral Lab 2026, 04/15-16/2026

Why this matters?

- BTF systems expose a large amount of real-time data through EPICS PVs
- Operators and beam line scientists need to quickly understand system state and react accordingly
- Raw data alone is not sufficient for safe and efficient operations
- Interfaces must support fast decision-making under time pressure

The problem

- EPICS exposes many PVs, often without structure from a user perspective
- Interfaces can become cluttered with too many values and no clear hierarchy
- Different systems may use inconsistent conventions and layouts
- Users are forced to interpret raw values instead of meaningful states

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Instance Name	Status	PV Count	Connected	Event Rate	Data Rate (GB/day)	Engine write thread(s)	Max ETL(%)
appliance0	Working	1459	563	131.2	0.2	0.02	0.02

Mental model

MONITORING

Real Hardware

(power supply, diagnostic system, motors...)

- Produces physical signals and responds to control actions
- Not directly accessible to users or applications
- Requires a control layer to interface with software

IOCs

- Software layer interfacing with the hardware
- Read data and sends commands to the device
- Run inside a container managed by EPIK8s
- Translate hw-level signals into PVs
- Ensure safe and controlled interaction with the system

PVs

- EPICS abstraction presenting data and control points
- Expose system information (RB, states, alarms, ...)
- Allow control action (setpoints, commands)
- Contact between Control System and application

GUI

- Application used by operator and users
- Reads PV to display system state and values
- Write to PV to send commands and setpoints
- Trasform raw data in meaning full info
- Ensure clarity and usability

CONTROL

Recap on Pv types

- **Readback (RBV):** represents measured or actual values (ai, bi, waveform records)
- **Setpoint (SET):** represents desired or commanded values (ao, bo, stringout)
- **State (STATE):** indicates system status - e.g. ON/OFF (bi, mbbi)
- **Mode (MODE):** defines operational mode - e.g. LOCAL/REMOTE (mbbo, mbbi)
- **Alarm (ALARM):** indicates severity or abnormal conditions (**NOT separate PVs by default)**

A real BTF example

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PV Name	Status	Appliance	Connected?	Monitored?	Sampling period	Last event	Details	Quick chart
BTF2:daq:DAQ_BTF2:gun_status	Being archived	appliance0	true	true	1.0	Mar/24/2026 09:34:55 UTC		
BTF2:daq:DAQ_BTF2:mod_status	Being archived	appliance0	true	true	1.0	Mar/24/2026 09:34:55 UTC		
BTF2:daq:DAQ_BTF2:particle_status	Being archived	appliance0	true	true	1.0	Mar/24/2026 06:37:41 UTC		
BTF2:slow:CaloV:value	Being archived	appliance0	true	true	1.0	Mar/26/2026 17:09:16 UTC		
BTF2:slow:EH1_CaloV:value	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:37:01 UTC		
BTF2:slow:EH1_calo_autotrack:value	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:36:49 UTC		
BTF2:slow:EH2:ele_instant:value	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:36:59 UTC		
BTF2:slow:EH2_CaloV:value	Being archived	appliance0	true	true	1.0	Mar/24/2026 09:30:17 UTC		
BTF2:slow:EH2_N_ele_tot:value	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:37:01 UTC		
BTF2:slow:EH2_calo_autotrack:value	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:37:07 UTC		
BTF2:slow:FITPIX:XCenter	Being archived	appliance0	true	true	1.0	Mar/24/2026 09:56:37 UTC		
BTF2:slow:FITPIX:XSTD	Being archived	appliance0	true	true	1.0	Mar/24/2026 09:56:37 UTC		
BTF2:slow:FITPIX:YCenter	Being archived	appliance0	true	true	1.0	Mar/24/2026 09:56:37 UTC		
BTF2:slow:FITPIX:YSTD	Being archived	appliance0	true	true	1.0	Mar/24/2026 09:56:37 UTC		
BTF2:slow:FITPIX:mean_hit	Being archived	appliance0	true	true	1.0	Mar/24/2026 09:56:37 UTC		
BTF2:slow:FITPIX:mean_sum	Being archived	appliance0	true	true	1.0	Mar/24/2026 09:56:37 UTC		
BTF2:slow:GaussProbe:DHRTB102:Error_source	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:37:06 UTC		
BTF2:slow:GaussProbe:DHRTB102:GPBIP	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:37:13 UTC		
BTF2:slow:GaussProbe:DHRTB102:value	Being archived	appliance0	true	true	1.0	Never		
BTF2:slow:GaussProbe:DHSTB001:Error_source	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:37:03 UTC		
BTF2:slow:GaussProbe:DHSTB001:GPBIP	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:37:10 UTC		
BTF2:slow:GaussProbe:DHSTB001:value	Being archived	appliance0	true	true	1.0	Mar/26/2026 17:10:54 UTC		
BTF2:slow:GaussProbe:DHSTB002:Error_source	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:36:51 UTC		
BTF2:slow:GaussProbe:DHSTB002:GPBIP	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:37:09 UTC		
BTF2:slow:GaussProbe:DHSTB002:value	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:36:50 UTC		

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PV Name	Status	Appliance	Connected?	Monitored?	Sampling period	Last event	Details	Quick chart
BTF2:slow:VUGPADME:value	Being archived	appliance0	true	true	1.0	Mar/26/2026 17:10:53 UTC		
BTF2:slow:VUGTB003:error_code	Being archived	appliance0	true	true	1.0	Mar/23/2026 08:54:26 UTC		
BTF2:slow:VUGTB003:error_status	Being archived	appliance0	true	true	1.0	Never		
BTF2:slow:VUGTB003:status	Being archived	appliance0	true	true	1.0	Mar/23/2026 08:54:26 UTC		
BTF2:slow:VUGTB003:timestamp	Being archived	appliance0	true	true	1.0	Mar/26/2026 17:10:53 UTC		
BTF2:slow:VUGTB003:value	Being archived	appliance0	true	true	1.0	Mar/23/2026 08:54:26 UTC		
BTF2:slow:VUGTB004:error_code	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:37:11 UTC		
BTF2:slow:VUGTB004:error_status	Being archived	appliance0	true	true	1.0	Never		
BTF2:slow:VUGTB004:status	Being archived	appliance0	true	true	1.0	Mar/23/2026 05:50:17 UTC		
BTF2:slow:VUGTB004:timestamp	Being archived	appliance0	true	true	1.0	Mar/26/2026 17:10:53 UTC		
BTF2:slow:VUGTB004:value	Being archived	appliance0	true	true	1.0	Mar/23/2026 05:50:17 UTC		
BTF2:slow:VUGTB201:error_code	Being archived	appliance0	true	true	1.0	Mar/18/2026 07:37:13 UTC		
BTF2:slow:VUGTB201:error_status	Being archived	appliance0	true	true	1.0	Never		
BTF2:slow:VUGTB201:status	Being archived	appliance0	true	true	1.0	Mar/23/2026 08:39:06 UTC		
BTF2:slow:VUGTB201:timestamp	Being archived	appliance0	true	true	1.0	Mar/26/2026 17:10:53 UTC		
BTF2:slow:VUGTB201:value	Being archived	appliance0	true	true	1.0	Mar/23/2026 08:39:06 UTC		
BTF2:slow:VUGTB202:error_code	Being archived	appliance0	true	true	1.0	Mar/23/2026 09:07:12 UTC		
BTF2:slow:VUGTB202:error_status	Being archived	appliance0	true	true	1.0	Never		
BTF2:slow:VUGTB202:status	Being archived	appliance0	true	true	1.0	Mar/23/2026 09:07:12 UTC		
BTF2:slow:VUGTB202:timestamp	Being archived	appliance0	true	true	1.0	Mar/26/2026 17:10:53 UTC		
BTF2:slow:VUGTB202:value	Being archived	appliance0	true	true	1.0	Mar/23/2026 09:07:12 UTC		
BTF:MAG:FAST:PS:OutputCurrent	Being archived	appliance0	true	true	1.0	Mar/26/2026 09:44:27 UTC		
BTF:MAG:FAST:PS:OutputVoltage	Being archived	appliance0	true	true	1.0	Mar/26/2026 09:44:27 UTC		
BTF:MAG:FAST:PS:STATE_RB	Being archived	appliance0	true	true	1.0	Mar/26/2026 09:44:29 UTC		
BTF:TML:SLTTB001D:ACCL_RB	Being archived	appliance0	false	true	1.0	Never		

EPICS Archiver Appliance Version 2.1.2

What does an operator actually need?

What does an operator actually need?

- Immediate understanding of system status
- Visibility of key physical values
- Clear indication of alarms or abnormal conditions
- Safe and controlled way to send commands

Control Interface Design Principles

Fast Interpretation

Understandable in < 1 sec

Focus on Decisions

Show what matters

Reusable patterns

Same PV → same behavior

Standardized Structure

Predictable layout

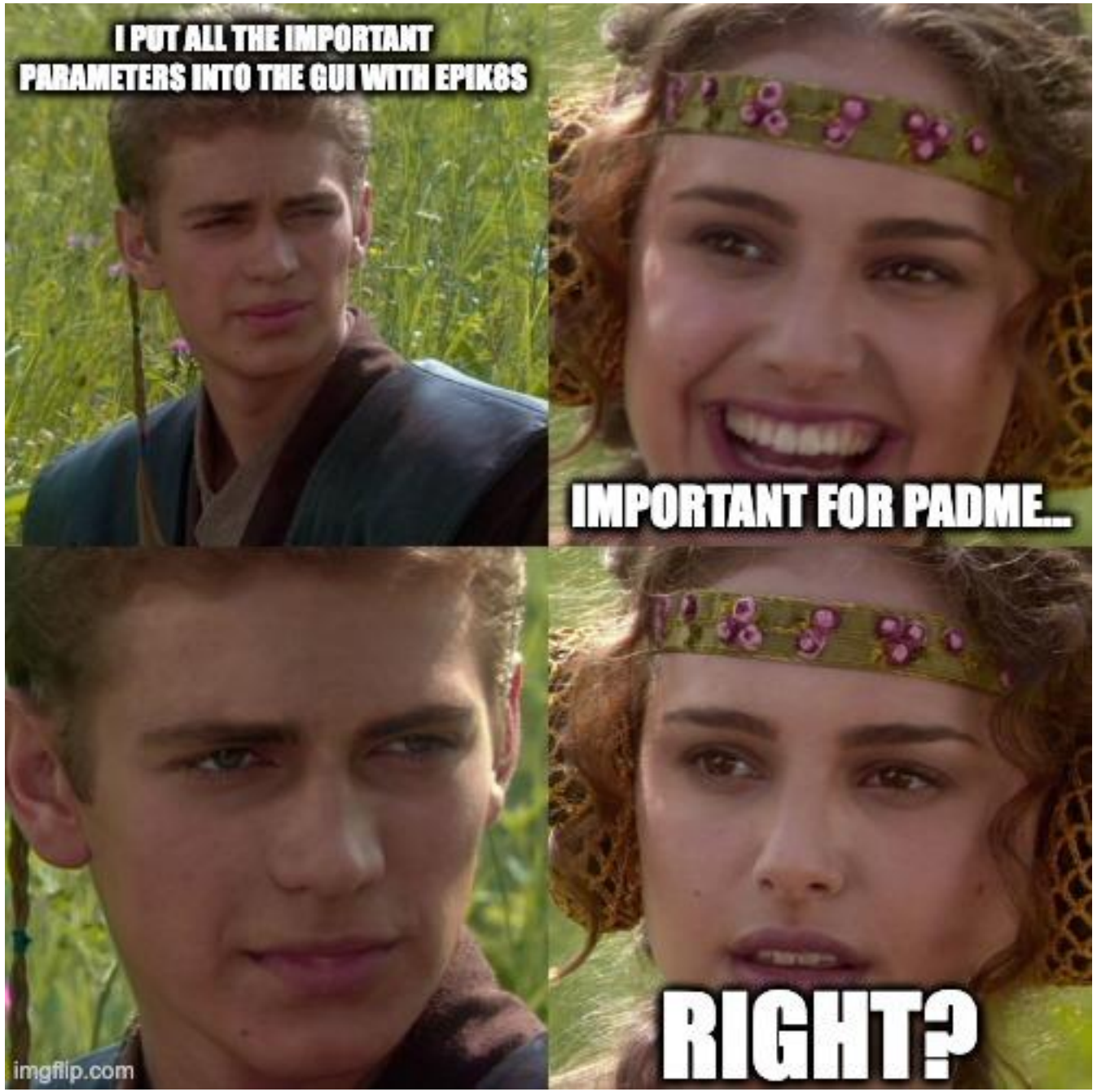
Some ideas...

- Monitor the Humidity, Pressure and Temperature of the EH1 and EH2
  - `BTF2:slow:PTUBTFEH1_Temp:value, BTF2:slow:PTUBTFEH1_Press:value, BTF2:slow:PTUBTFEH1_RH:value`
  - `BTF2:slow:PTUBTFEH2_Temp:value, BTF2:slow:PTUBTFEH2_Press:value, BTF2:slow:PTUBTFEH2_RH:value`
- Monitor the values of the Hall probes installed on the magnet
  - `BTF2:slow:GaussProbe:DHSTB001:value, BTF2:slow:GaussProbe:DHSTB201:value, BTF2:slow:GaussProbe:DHSTB202:value`
- Monitor the vacuum level
  - `BTF2:slow:VUGTB003:value, BTF2:slow:VUGTB004:value, BTF2:slow:VUGTB201:value, BTF2:slow:VUGTB202:value, BTF2:slow:VUGPADME:value`
- Monitor the current of the the magnets
  - DAFNE:BTF:QUATB101:current_rb, DAFNE:BTF:QUATB102:current_rb, (...)
  - DAFNE:BTF:DHSTB001:current_rb, DAFNE:BTF:DHPTB101:current_rb, (...)
  - DAFNE:BTF:CHHTB001:current_rb, DAFNE:BTF:CVVTB001:current_rb, (...)

The final task...

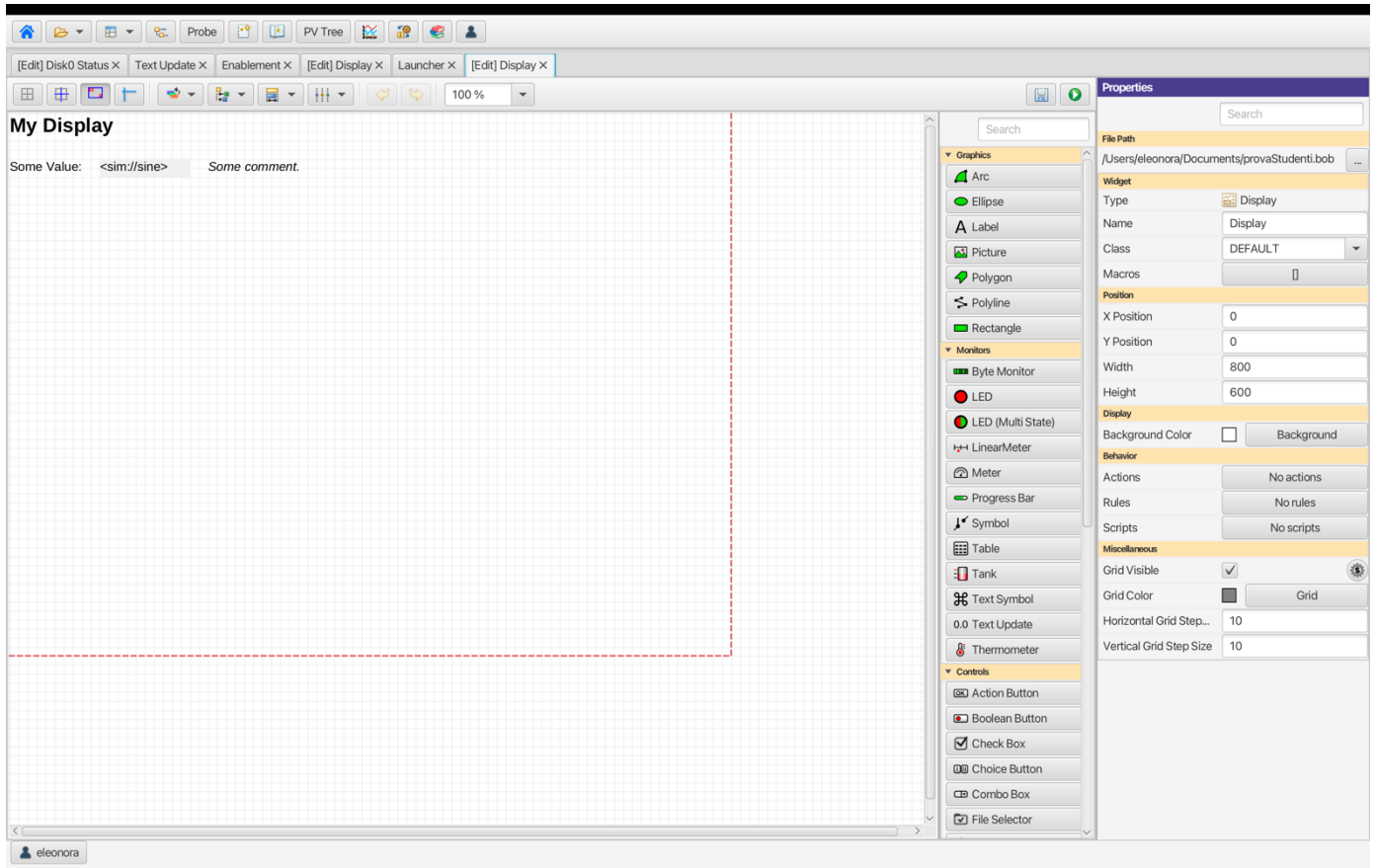
Let's realize together a GUI to show all the pivotal PV needed by PADME

THINK ABOUT WHAT USER WANT TO SEE



The exercise…

1. Check if the PV is archived on the EPICS archiver
2. Check the graph to understand the range of the PV
3. Open a new Display on Phoebus



- Use text update to see the value of a given PV changing (eventually) over the time
- Use LED, Stripcharts, … to give a graphical representation of the PV
- Open the terminal phoebus -s settings.ini –resource DoctoLab.bob